Causal-RAG

Counterfactual-Audited Retrieval for High-Stakes Lending Decisions

Complete project report: the idea, the method, every real-data experiment, the problems we hit, the solutions we built, and the final results

Project	Causal-RAG: four cooperating agents (Retriever, Causal Attribution, Counterfactual Perturbation, Synthesis & Audit)
Built on	CALF: a tree-structured causal data fabric for individual treatment effect estimation (the author's own paper)
Data	Real only: Lending Club accepted loans (2012-2015), Fannie Mae and CFPB policy documents, the Hillstrom randomised e-mail experiment. No synthetic data anywhere in the pipeline.
Compute	Kaggle notebooks (GPU T4), Qwen2.5-7B-Instruct (4-bit) as the language model
Code version	2.0.0 (final). Seven archived runs: v1.0, v1.1 (two sessions), v1.2, v1.3, v1.4, v2.0
Report date	18 October 2026

Contents

1. Executive summary **4**

2. Background and project goal **5**

2.1 The original idea . . . 5

2.2 CALF in brief . . . 6

2.3 Ground rules agreed at the start . . . 6

2.4 Research questions . . . 6

3. Data **6**

3.1 Lending Club accepted loans (the main dataset) . . . 6

3.2 Policy documents (real regulatory text) . . . 7

3.3 Hillstrom e-mail experiment (randomised validation data) . . . 7

4. Method: how the system was built **7**

4.1 Mapping CALF's layers onto a loan application . . . 7

4.2 The four underwriting clauses (treatments) . . . 7

4.3 Structural causal graph and adjustment sets . . . 8

4.4 Adapting CALF to real data . . . 9

4.5 Evaluating effects without ground truth . . . 9

4.6 The Retriever Agent . . . 10

4.7 Language-model scoring and the prompt systems . . . 10

4.8 The Counterfactual Perturbation Agent . . . 10

4.9 The Synthesis & Audit Agent . . . 11

4.10 The local cliff test . . . 11

4.11 Software, packaging and compute . . . 11

5. Results **12**

5.1 CALF on real loans (RQ1) . . . 12

5.2 Validation on a real randomised experiment . . . 13

5.3 Refutations and E-values on the observational effects . . . 13

5.4 Retriever evidence quality (before and after the fix) . . . 13

5.5 Do language-model systems benefit from causal evidence? (RQ2) . . . 14

5.6 Is the LLM consistent with the causal estimates? (RQ3) . . . 16

5.7 The audit agent (RQ4) . . . 16

5.8 The cliff test: does the LLM react to the cutoff itself? . . . 17

5.9 Band tables used in the final experiment . . . 18

6. Problems we met and how we solved them **19**

7. Run history **20**

8. Conclusions **20**

8.1 Answers to the research questions . . . 20

8.2 What the project contributes . . . 21

8.3 Limitations 21
8.4 Suggested future work 21
9. How to reproduce and use the project **22**
Appendix A. Glossary **23**
Appendix B. Observations on the CALF paper **23**
Appendix C. Prompt examples **23**

1. Executive summary

What we set out to do. Standard retrieval-augmented generation (RAG) finds documents that *look similar*, but similarity is not causation: in underwriting, similar past cases repeat past biases and mistake confounders for real risk drivers. The goal was to build a retrieval system whose evidence is *causally adjusted*, whose language-model answers are *stress-tested with counterfactuals*, and whose final output is an *auditable decision with a causal graph* rather than a wall of text, using the author's own CALF model for the causal estimation and using only real data.

What we built. A complete, reproducible Kaggle project (Python package plus notebooks) with four agents: a **Retriever** (dense retrieval plus a structural causal graph and balance-space matched precedent), a **Causal Attribution Agent** (CALF, estimating the effect of each underwriting clause on default), a **Counterfactual Perturbation Agent** (edits the applicant or the evidence and checks whether the language model reacts the way the causal estimate says it should) and a **Synthesis & Audit Agent** (APPROVE / DECLINE / REFER_TO_HUMAN with a causal DAG, JSON record and written rationale). On top of this we built validation tools suited to real data (no ground-truth individual effects exist): doubly-robust calibration tests, a real randomised experiment, refutation tests, E-values, control experiments and a local "cliff" test. A live Gradio demo is included.

Headline findings

- **CALF works on real loans.** Its predicted effect heterogeneity is well calibrated for three of the four underwriting clauses (calibration slope 0.86-1.18 where 1 is ideal; utilization is weak at 0.19), while a LightGBM T-learner is badly over-dispersed (slopes 0.04-0.27).
- **It was validated against a real randomised experiment** (Hillstrom): the model ATE of 0.066-0.069 lies inside the experimental interval 0.059 [0.045, 0.073], and CALF ranks customers far better than a T-learner (Qini about 31 vs 9).
- **All refutation tests passed** (placebo treatment, random common cause, subsample) for all four clauses; E-values are 1.8-2.1 for DTI, FICO and loan-to-income, and fragile (1.06) for utilization.
- **No language-model system was clearly better than the plain language model.** On 600 held-out 2015 applicants (85 defaults) AUROC was 0.633 for the plain LLM and 0.658-0.678 for the Causal-RAG variants (differences +0.02 to +0.04, p = 0.06 at best). Tabular models were at least as good as every LLM variant (CALF risk 0.707, LightGBM 0.691).
- **A control that only lists which risk flags apply matched the full causal pipeline** (AUROC 0.668 vs 0.678, p = 0.59), so the causal evidence added no measurable accuracy beyond the flags.
- **Prompts that describe conditions as cutoffs make the model step-like.** A local cliff test shows plain reasoning is smooth (index 1.08) and a prompt without cutoff wording is smooth (0.73), while the flag-label prompt jumps 86 times more across the cutoff than away from it. The graded prompt reduces the cliff to 2.8 (2.3 with band tables) but does not remove it.
- **The audit machinery is the reusable contribution.** It exposed that the apparent improvement was a flag-listing effect, that agreement with the causal sign is an easy test, and that some prompt designs behave like hard rules.

> **Bottom line.** Causal effect estimation on real data works and can be validated honestly. Feeding those estimates to an LLM did not measurably improve its lending decisions, and describing conditions as cutoffs makes its behaviour step-like. Everything in this report is reported as measured, including the negative results and the mistakes we found and corrected in our own metrics.

Key numbers at a glance

Item	Value
Loans used (36-month, 2012-2015, resolved)	589,099 (train 260,136 / validation 46,076 / test 2015: 282,887)
Underwriting clauses analysed	4 (DTI > 25%, utilization > 75%, FICO < 680, loan-to-income > 20%)
CALF calibration slope (1 = ideal)	FICO 0.92, loan-to-income 1.18, DTI 0.86, utilization 0.19

Item	Value
Hillstrom RCT: model ATE vs experiment	0.066-0.069 vs 0.059 [0.045, 0.073]
AUROC, 600 applicants: plain LLM / best Causal-RAG / CALF risk / LightGBM	0.633 / 0.678 / 0.707 / 0.691
Flags-only control vs full Causal-RAG	0.668 vs 0.678 (p = 0.59)
Cliff index (1 = smooth): plain / bands-only / graded / flags-only / smooth-label	1.08 / 0.73 / 2.76 / 5.29 / 86.0
Audit: default rate approved / referred / declined (v1.2, n = 600)	11.5% / 17.7% / about 23%
LLM scoring runs (GPU T4)	v1.0 83 min, v1.1 94 min, v1.2 351 min, v1.3 295 min, v1.4 125 min, v2.0 143 min

2. Background and project goal

2.1 The original idea

The project started from this pitch: *Standard semantic search retrieves documents that match lexically and topically, but fails at causal reasoning. In automated underwriting, medical protocols or policy compliance, matching similar historical cases often repeats past biases or mistakes confounding variables for true risk drivers.* The proposed architecture had four agents:

- **Retriever Agent**: a dense retriever paired with a structural causal graph over document metadata and key entities.
- **Causal Attribution Agent**: adapts the CALF project to estimate the individual treatment effect of specific policy clauses on the outcome.
- **Counterfactual Perturbation Agent**: generates minimal counterfactual perturbations of the retrieved evidence and checks whether the LLM's recommendation flips appropriately, as a causal sanity check.
- **Synthesis & Audit Agent**: emits the final decision together with an auditable causal DAG instead of only generated text.

The motivation is the compliance and interpretability problem that banks and healthcare enterprises face when deploying LLMs, and it directly showcases the author's causal machine-learning work.

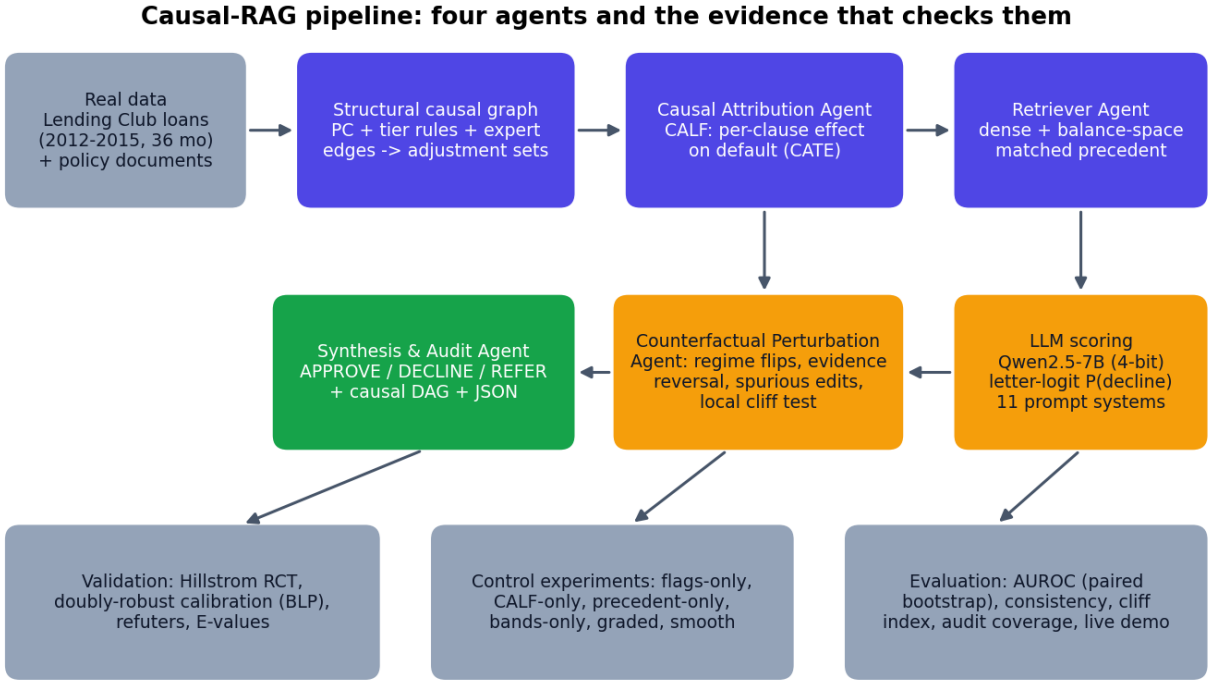


Figure 1. The four agents, the causal graph that feeds them, and the evidence (validation, controls, evaluation) used to check them.

2.2 CALF in brief

CALF (Causally-Aware Layered Fabric) is a tree-structured neural architecture for individual treatment effect (ITE) estimation. It splits covariates into four semantic groups: **Roots** (fixed background), **Soil** (environment), **Leaves** (social/peer context) and **Choice** (the unit's own decisions); encodes each group separately; passes information along the assumed causal order Roots → Soil → Leaves → Choice with multi-head cross-attention; fuses the four representations; and predicts a propensity score and two potential outcomes. It is trained with an outcome loss, a propensity loss and a soft monotonicity prior. The paper proves nonparametric identification under strong ignorability and overlap, evaluates on an NLSY97-derived semi-synthetic benchmark (lowest PEHE of six methods), and reports its limitations candidly: sample-hungry, miscalibrated MC-Dropout intervals, and a **null result in insurance underwriting** where the layered structure gave no benefit. That last point mattered for this project: the paper itself says CALF's advantage appears only when the covariates truly decompose into causal layers.

2.3 Ground rules agreed at the start

- **No synthetic data.** Everything must run on real datasets available on Kaggle or downloadable from official sources.
- **Kaggle notebooks** as the compute platform (GPU T4, 30 GB RAM, session limits), so the project had to be resumable and package-based.
- **Honest evaluation.** Real observational data contain no ground-truth individual effects, so validity must be argued with several independent kinds of evidence: (1) an estimator validated on real randomised trials, (2) refutation and sensitivity tests on the observational data, (3) calibration tests against doubly-robust pseudo-outcomes, and (4) explicit statements of selection bias and untestable assumptions.

2.4 Research questions

- **RQ1.** Can CALF credibly estimate the effect of individual underwriting clauses on loan default using real data?
- **RQ2.** Does giving an LLM causally adjusted evidence (matched precedent plus CALF effects) improve its lending decisions compared with plain prompting, vanilla RAG and kNN RAG?
- **RQ3.** Is the LLM's behaviour consistent with the causal estimates when the applicant or the evidence is perturbed counterfactually?
- **RQ4.** Can the combination be turned into an auditable decision record (decision, flags, causal DAG) that a human reviewer can contest?

3. Data

3.1 Lending Club accepted loans (the main dataset)

We used the Kaggle dataset *wordsforthewise/lending-club* (accepted loans, 2007-2018). It is real, large, contains real default outcomes, has a time dimension for honest splits, and matches the banking story of the pitch. The preparation pipeline (`data_prep.py`) makes these deliberate choices:

- **Only pre-origination fields** (known at application time): income, employment length, home ownership, verification status, purpose, state, DTI, delinquencies, inquiries, open/total accounts, public records, revolving balance and utilization, mortgage accounts, credit-history length, FICO, loan amount, employer and loan title. No payment-history or recovery columns (post-outcome leakage).
- **The lender's own risk outputs were excluded**: *grade*, *sub_grade* and *int_rate* are the previous underwriting decision (mediators), and using them as causes would leak the old decision.
- **36-month loans issued 2012-2015** with a resolved outcome (Fully Paid = 0, Charged Off = 1). Loans flagged "does not meet the credit policy" were excluded.

- **Right-censoring guard.** For each 2015 issue month we measured the share of loans still unresolved. It never exceeded 0.32% (December 2015), so no months had to be cut; the check is part of the pipeline and would have trimmed the test cohort if the share had exceeded 1%.
- **Temporal split.** Train = 2012-2014 (85% of it), validation = a random 15% of 2012-2014, test = all of 2015, so the test set is strictly later than anything the models or retrieval index have seen.

Role	Loans	Default rate
train (2012-2014)	260,136	0.1325
validation (random 15% of 2012-2014)	46,076	0.1320
test (2015)	282,887	0.1488

A 306,212-loan retrieval index was built from train + validation loans (each loan rendered as a text "card" and embedded). An example card (real loan):

```
Applicant: annual income $35,000; employed unknown years; home ownership MORTGAGE; state TX; income verification: Source Ve
Credit profile: FICO 677; debt-to-income 12.9%; revolving utilization 93%; revolving balance $15,006; delinquencies (2y) 0;
inquiries (6m) 0; open accounts 14; total accounts 27; public records 0; mortgage accounts unknown; credit history 10.4 yea
Request: $12,000 for other (loan-to-income 34.3%).
```

3.2 Policy documents (real regulatory text)

Four public documents were downloaded from their official sites and chunked into **88 passages**: Fannie Mae Selling Guide *B3-6-02 Debt-to-Income Ratios* and *B3-5.1-01 General Requirements for Credit Scores*; CFPB / eCFR *12 CFR 1026.43* (ability to repay) and *12 CFR 1002.6* (Regulation B, evaluation of applications). Passages were tagged with entities (income 26, repayment 18, DTI 17, credit score 13, utilization 4, prohibited basis 4). Users can add their own documents through a `policy_docs/` folder.

3.3 Hillstrom e-mail experiment (randomised validation data)

The Hillstrom MineThatData experiment randomly assigned customers to a men's e-mail, a women's e-mail or no e-mail; we used "any e-mail vs none" as the treatment (propensity known, about 2/3) and website visit as the outcome. Because assignment was random, the experimental difference in means is a trustworthy reference to test whether an estimator recovers true average and heterogeneous effects. The Criteo uplift dataset was planned as a second experiment but its download link (used by the scikit-uplift helper) now returns HTTP 404, so it was skipped.

4. Method: how the system was built

4.1 Mapping CALF's layers onto a loan application

CALF's four layers were mapped to the temporal/causal tiers of a loan application: things fixed about the person come first, then environment, then the credit profile, then the request. A treatment lives in one tier; its **adjustment set** is every covariate in strictly earlier tiers plus same-tier covariates that are not part of, or mechanically tied to, the treatment. This is exactly the paper's ignorability assumption (outcomes independent of treatment given Roots, Soil and Leaves) generalised so a treatment can sit in any tier.

CALF layer / tier	Columns
0 Roots	log annual income, employment years (+ missing flag), credit-history length, home ownership
1 Soil	state, income-verification status (a lender-process variable; flagged as a judgment call)
2 Leaves	FICO, delinquencies (2y), inquiries (6m), open accounts, public records, total accounts, mortgage accounts, revolving balance (log), revolving utilization, DTI
3 Choice	log loan amount, loan-to-income, loan purpose
4 Outcome	default (charged off)

4.2 The four underwriting clauses (treatments)

Each clause is a binary "regime". The effect we estimate is the individual effect on default probability of being in the regime rather than outside it, *for the same background profile*. The columns that define the treatment (and anything mechanically tied to it) are removed from the inputs; otherwise the treatment would be a deterministic function of the covariates and overlap would be destroyed.

Key	Clause	Tier	Dropped from inputs	Treated share	Cutoff origin
dti_gt25	DTI above 25%	2	dti	0.205	analyst-defined
util_gt75	Revolving utilization above 75%	2	revol_util	0.214	analyst-defined
fico_lt680	FICO below 680	2	fico	0.361	analyst-defined (Fannie Mae minimum is 620)
lti_gt20	Loan amount above 20% of income	3	lti, loan_amnt_log	0.427	analyst-defined

A design mistake we found and fixed. The first version used Fannie Mae's 36% DTI guideline. In real Lending Club data only **1.5%** of loans exceed it, so 85% of test loans had a propensity below 0.05 (a positivity failure) and the effect was not identifiable. The clause became `dti_gt25`, an honestly analyst-defined cutoff with 20.5% treated share. The 36% finding itself is a result: the policy cutoff binds on almost nobody in this population.

4.3 Structural causal graph and adjustment sets

The graph is built with the standard "discovery plus knowledge" recipe: (1) the PC algorithm (causal-learn, Fisher-z tests) on a 30,000-loan sample of numeric variables; (2) **temporal-tier background knowledge**: no edge may point from a later tier to an earlier tier (a loan request cannot cause a person's credit age), which orients every cross-tier edge; (3) **expert and mechanical edges** for definitional relations (loan-to-income = amount / income; utilization = balance / limit; FICO components) and for the four clause variables pointing to default; experts override discovered orientations. Categorical variables (state, purpose) enter as expert-only nodes.

Two adjustment sets are produced per clause: **A_default** (tier rule only) and **A_graph**, which additionally removes the directed descendants of the treatment variable and its direct undirected neighbours (possible mediators or colliders, so-called "bad controls"). For example, for utilization the graph says that FICO, DTI, inquiries, open accounts, public records and loan-to-income are downstream, so A_graph drops them. CALF is fitted on both and the difference is reported as a sensitivity analysis (Section 5.1).

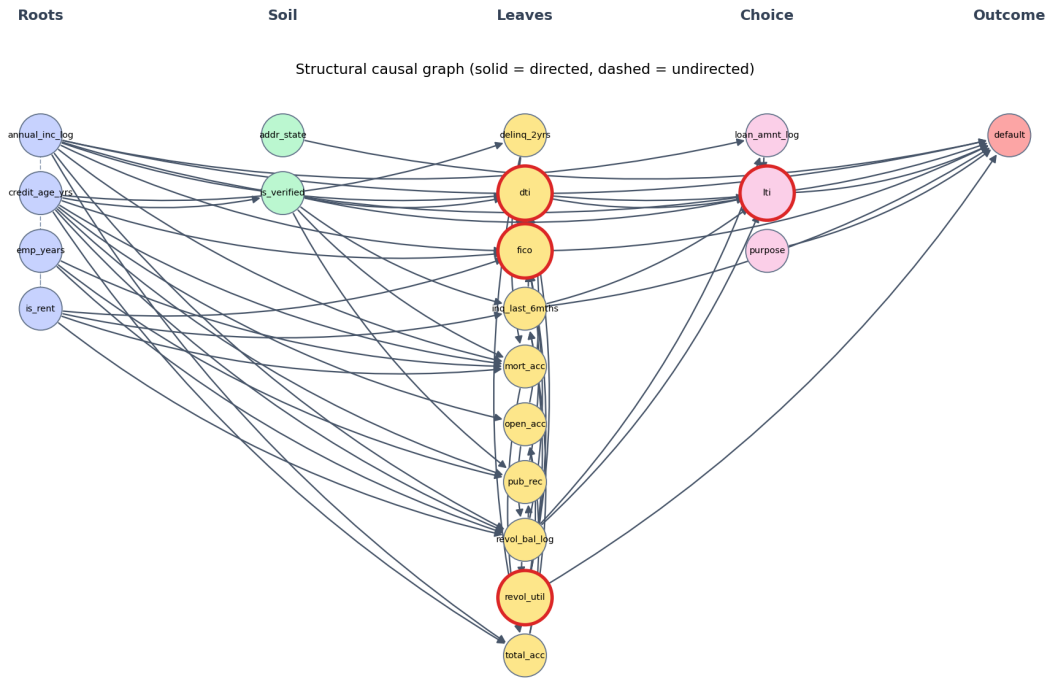


Figure 2. The discovered/expert causal graph, laid out by tier (Roots, Soil, Leaves, Choice, Outcome). Solid arrows are directed edges, dashed lines are undirected associations.

4.4 Adapting CALF to real data

The paper's CALF was built for a simulated continuous outcome with known individual effects. Real data forced these deliberate changes:

Aspect	Change and reason
Outcome	Binary default: cross-entropy on logits; effect = p1 - p0 (a risk difference).
Model selection	PEHE needs true individual effects (never observed). Checkpoints are selected on MSE against a cross-fitted doubly-robust pseudo-outcome.
Monotonicity prior	The paper's soft prior (beta = 0.5) is set to 0 by default: baking in "effects are non-negative" would make the counterfactual audit partly circular.
Layer count	A variable number of layers (tiers 0 to the treatment tier), since a clause can only be adjusted for by upstream covariates.
Attention	In the paper each layer is one vector, so cross-attention has a single key and the softmax is identically 1: it reduces to a linear map. We added k_tokens (each layer split into 4 latent tokens) so attention can select; k_tokens = 1 reproduces the paper as written and is run as an ablation.
Training	Batch 2,048 and up to 40 epochs with early stopping (the paper used batch 64 and 200 epochs on 9k units; here n is about 260k). Ensemble of 3 seeds; between-seed spread replaces MC-Dropout, which the paper itself shows under-covers (66.9% vs 90%).
What the LLM may see	CALF's effect estimates, never its factual risk prediction, so AUROC differences cannot be CALF's risk leaking through the prompt. CALF risk and LightGBM are reported as separate reference rows.

4.5 Evaluating effects without ground truth

Because no individual treatment effect is ever observed in real data, four kinds of evidence are used together:

- **Doubly-robust (AIPW) pseudo-outcome** tau_dr = mu1 - mu0 + (T - e) / (e (1 - e)) (y - mu_T), with nuisances cross-fitted (LightGBM) inside each evaluation split. Its conditional mean equals the true effect if either the propensity or the outcome model is right, so the squared error against it ranks models like PEHE would (absolute values are not comparable to a paper PEHE).
- **Best-linear-predictor calibration (BLP) slope**: regress tau_dr on the predicted effect. 1 = predicted heterogeneity is real and correctly scaled; 0 = no signal; above 1 = under-dispersed; between 0 and 1 = over-dispersed. Quintile group-effect tables (GATES) show the same visually.

- **Real randomised experiment** (Hillstrom): known propensity, experimental difference in means as the reference, Qini coefficient for ranking.
- **Refutations and sensitivity** on the observational estimates: placebo treatment (permute T, effect must vanish), random common cause (add noise covariate, estimate must not move), 80% subsets (stability) and E-values (how strong an unmeasured confounder would have to be to explain the effect away).

4.6 The Retriever Agent

For each applicant and each clause the retriever assembles an evidence bundle in four steps:

- **Dense stage.** Text cards are embedded with *BAAI/bge-small-en-v1.5*; the top 2,000 historical loans by cosine similarity form a candidate pool (only labelled, pre-2015 loans).
- **Causal stage.** Candidates are matched like-for-like in a two-dimensional **balance space**: logit propensity e_k and logit prognostic risk p0_k from CALF's heads, computed on the graph-defined adjustment set (propensity/prognostic-score matching). The 25 nearest treated and 25 nearest control loans are selected. (Version 1.1 added the nearest treated/control loans across the whole index to the candidate pool, because a text-similarity pool alone left the matches badly unbalanced; see Section 5.4.)
- **Local effect.** The difference in default rates between matched arms with a confidence interval, the standardised mean difference (SMD) of the balance scores, and a "thin evidence" flag when either arm has fewer than 12 loans or SMD exceeds 0.5.
- **Policy stage.** Policy text is retrieved by dense similarity restricted to passages whose entity tags link to the clause's variable (the graph's entity layer). A clause is called *policy-supported* only if the retrieved text actually states its cutoff (regex on the cutoff number); otherwise it is labelled *topical only / analyst-defined*. Passages tagged with the parent variables in the graph are kept as context for the audit trail.

4.7 Language-model scoring and the prompt systems

The language model is **Qwen2.5-7B-Instruct** in 4-bit NF4 (Kaggle T4). Decisions are not parsed from free text: the prompt ends with the question "A = approve, B = decline, answer with one letter" and the score is the next-token logit difference margin = logit(B) - logit(A), a continuous value (deterministic, usable for AUROC, calibration and counterfactual deltas, immune to formatting failures). Free text is generated only for the audit rationale. Eleven prompt systems were compared, all on the same applicants:

System	What the model sees
no_rag	Applicant card only (plain LLM; the reference).
vanilla_rag	Card + top policy passages + 6 most text-similar past loans with outcomes.
knn_rag	Card + 6 nearest past loans in feature space, with outcomes.
causal_rag (v1.1 prompt)	For each clause that APPLIES: policy basis, matched precedent (treated vs control counts), CALF effect with spread. Clauses that do not apply are not shown.
causal_rag_smooth	All four clauses always shown in the same layout, with the applicant's value and a TRIGGERED / not triggered label, plus a sentence that cutoffs are conventions, not cliffs.
causal_calf_only	Control: CALF effect numbers, no matched precedent.
causal_precedent_only	Control: matched precedent, no CALF effect numbers.
causal_conditions_only	Control: only the list of applicable standard risk flags (no policy, no precedent, no numbers).
causal_graded	Like smooth but without the TRIGGERED word: each clause shows the applicant's value and its signed distance to the cutoff.
bands_only	Control: applicant + unadjusted historical default rate by value band for DTI, utilization, FICO and loan-to-income. No cutoffs, flags, regimes, precedent or CALF numbers.
causal_graded_bands	The graded prompt plus the same band tables.

4.8 The Counterfactual Perturbation Agent

The agent edits the applicant or the evidence, re-asks the same LLM with the same template, and compares the change in P(decline) with CALF's effect. Families of perturbation:

- **Regime flip**, do(T to 1 - T): the clause field moves to the *median value of the opposite arm of the matched precedent*. This is the contrast CALF's estimate actually describes ("same background, other regime"), so the expected sign of the LLM's change is direction times sign of CALF's effect.
- **Boundary flip**: the field is moved just across the cutoff. (We later found this measures something different from what we first thought; see Section 5.8 and Problem 15.)
- **Evidence reversal** (swap the matched-precedent counts between arms) and **evidence removal**: a model that really uses the evidence must move opposite to the original local effect when the evidence is reversed.
- **Spurious edits**: state, employer title and loan title (fields no clause references) are replaced by real random values; the decision should not move.
- **Local cliff test** (v1.4): see 4.10.

4.9 The Synthesis & Audit Agent

The decision rule is deterministic given its inputs; the LLM supplies only the margin and a rationale. The base decision is DECLINE if the margin exceeds a label-free threshold (the riskiest 20% of the sample) and APPROVE otherwise. Five counted audit flags are computed from the tests above: *llm_calf_disagree* (LLM moved the opposite way to CALF's sign on a regime flip), *evidence_model_conflict* (matched precedent and CALF disagree in sign, CI excluding 0), *thin_overlap* (thin or unbalanced evidence, or propensity outside 0.05-0.95), *evidence_ignored* (LLM moved the wrong way when the evidence was reversed) and *spurious_sensitive* (decision changes when only irrelevant fields change). *policy_unsupported* is informational only. The decision becomes **REFER_TO_HUMAN** if at least two flags are raised, or one flag with a margin within a small band of the threshold. Each case produces a JSON record (decision, flags, per-clause table, evidence case ids, input check for prohibited bases), a causal DAG image with CALF effects on the clause-to-default edges coloured by whether the LLM agreed, a self-contained HTML report and an LLM-written rationale that is clearly separated from the audit table.

4.10 The local cliff test

For each clause and applicant (all other fields unchanged) the LLM is queried at four values of the clause field: just below and just above the cutoff (thr - d and thr + d) and an equally wide interval far from the cutoff (thr - 5d and thr - 3d, the placebo). **cliff index = mean |change in P(decline) across the cutoff| / mean |change over the placebo interval|.** A model reasoning smoothly about the number changes about as much in both (index about 1); a model reacting to a label or regime changes far more across the cutoff (index much greater than 1). Confidence intervals are bootstrap over applicants.

4.11 Software, packaging and compute

- **Python package** `causal_rag` (about 20 modules): data_prep, config (tiers, clauses, featurizer), causal_graph, policy_corpus, embedding, retrieval, calf, dr_eval, refute, rct_validation, prompts, llm, counterfactual, audit, eval, pipeline, compare, live_demo, report, paths.
- **Notebooks**: stage notebooks 01-08, then the experiment notebooks (v1.2, v1.3, v1.4) and the final v2.0 notebook; all keep results in an `artifacts/` folder that can be saved as a Kaggle dataset, and all long stages are chunked and resumable.
- **Run history** is stored in `results/<run>/` folders inside the code dataset; every new run compares itself with the archived ones.
- **Libraries**: PyTorch, LightGBM, causal-learn, sentence-transformers, bitsandbytes, Gradio, pandas, scikit-learn; EconML was used for a causal-forest baseline in an early run.
- **Live demo**: Gradio app with an audit tab (decision, flags, DAG, per-clause table) and a what-if sweep tab that plots P(decline) against a continuous field for several prompt designs.

5. Results

5.1 CALF on real loans (RQ1)

Table 5.1. CALF and baselines on the 2015 test cohort (v1.1 run). DR-MSE is against the cross-fitted doubly-robust pseudo-outcome; BLP slope 1 = ideal; ATE (DR) is the doubly-robust average effect on the trimmed overlap region (identical across models by construction); outside overlap = share of test loans with estimated propensity outside [0.05, 0.95] (excluded from the metrics).

Clause	Model	DR-MSE	BLP slope	slope SE	ATE (DR)	outside overlap
dti_gt25	CALF layered (K=4) [primary]	1.0120	0.8557	0.1304	0.0416	0.1752
dti_gt25	CALF K=1 (paper attention)	1.0122	0.5272	0.0916	0.0416	0.1752
dti_gt25	Flat MLP	1.0123	0.9365	0.1472	0.0416	0.1752
dti_gt25	LightGBM T-learner	1.0143	0.1880	0.0407	0.0416	0.1752
dti_gt25	CALF K=4, graph-derived set	1.0122	0.5363	0.1042	0.0416	0.1752
util_gt75	CALF layered (K=4) [primary]	1.0696	0.1892	0.2437	0.0129	0.2953
util_gt75	CALF K=1 (paper attention)	1.0696	0.4406	0.2439	0.0129	0.2953
util_gt75	Flat MLP	1.0695	0.4198	0.2261	0.0129	0.2953
util_gt75	LightGBM T-learner	1.0719	0.0369	0.0512	0.0129	0.2953
util_gt75	CALF K=4, graph-derived set	1.4283	0.3944	0.2306	0.0173	0.2953
fico_lt680	CALF layered (K=4) [primary]	0.8253	0.9166	0.1111	0.0405	0.0826
fico_lt680	CALF K=1 (paper attention)	0.8253	0.7158	0.0841	0.0405	0.0826
fico_lt680	Flat MLP	0.8252	1.0905	0.1176	0.0405	0.0826
fico_lt680	LightGBM T-learner	0.8273	0.1552	0.0397	0.0405	0.0826
fico_lt680	CALF K=4, graph-derived set	0.8255	0.5648	0.0864	0.0405	0.0826
lti_gt20	CALF layered (K=4) [primary]	0.7704	1.1799	0.0995	0.0403	0.0610
lti_gt20	CALF K=1 (paper attention)	0.7703	1.4733	0.1079	0.0403	0.0610
lti_gt20	Flat MLP	0.7702	1.2254	0.0903	0.0403	0.0610
lti_gt20	LightGBM T-learner	0.7718	0.2702	0.0405	0.0403	0.0610
lti_gt20	CALF K=4, graph-derived set	0.7702	0.9382	0.0776	0.0403	0.0610

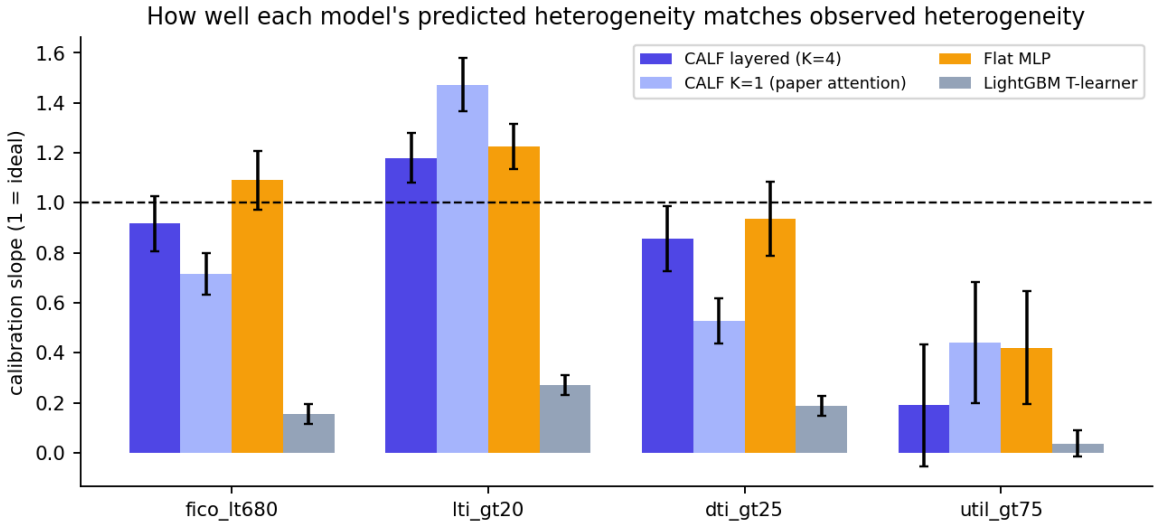


Figure 3. Calibration slope of predicted heterogeneity by clause and model (error bars = standard error; dashed line = ideal 1).

- **Calibration.** The layered CALF is close to ideal for FICO (0.92) and loan-to-income (1.18), good for DTI (0.86) and weak for utilization (0.19, standard error 0.24: no usable heterogeneity signal for that clause).
- **Layering did not help.** CALF with 4 attention tokens, with 1 token (the paper as written) and a flat MLP have the same DR-MSE to three decimals for every clause. The slopes differ somewhat (e.g.

K=1 gives 0.53 for DTI, 1.47 for loan-to-income) but within noise of each other. This repeats the paper's insurance finding: the tree structure pays off only when covariates truly decompose into causal layers.

- **The T-learner is over-dispersed.** LightGBM T-learner slopes are 0.04-0.27: it produces far more heterogeneity than the data support. CALF and the flat MLP are much better calibrated.
- **Adjustment-set sensitivity.** Using the graph-derived set (dropping possible mediators and bad controls) leaves the average effects nearly unchanged (DTI 0.042, FICO 0.041, loan-to-income 0.040) but lowers the calibration slopes for DTI (0.54) and FICO (0.56); for utilization, where the graph removes eight other variables including FICO, DTI and revolving balance, the average effect rises from 0.013 to 0.017. Conclusions about heterogeneity therefore depend somewhat on which variables are controlled, which is why both sets are reported.
- **The first version failed on DTI.** With the 36% cutoff, CALF's slope was -0.16 (no signal) and 85% of loans fell outside overlap; this is what led to the dti_gt25 redesign (Section 4.2).

5.2 Validation on a real randomised experiment

Table 5.2. Hillstrom randomised e-mail experiment (15% held-out customers; known propensity). Qini is on the count scale (higher = better ranking of who benefits).

Model	Model ATE	Experiment ATE	Experiment 95% CI	inside CI	BLP slope	slope SE	Qini
CALF K=4 tokens (layered)	0.0686	0.0588	[0.0449, 0.0728]	yes	0.9040	0.3216	30.6708
CALF K=1 (paper-as-written)	0.0674	0.0588	[0.0449, 0.0728]	yes	1.0602	0.3340	31.3096
CALF flat (1 layer)	0.0657	0.0588	[0.0449, 0.0728]	yes	1.0938	0.3752	31.7323
LightGBM T-learner	0.0592	0.0588	[0.0449, 0.0728]	yes	0.1234	0.1262	9.3222

- All models recover the average effect (the T-learner is closest, 0.0592 vs 0.0588), but only the neural models rank customers by benefit: Qini 30.7-31.7 against 9.3, and calibration slopes 0.90-1.09 against 0.12.
- As on Lending Club, the layered, K=1 and flat variants are statistically indistinguishable here, where the covariates have no strong layer structure.
- This is the independent evidence that the estimator recovers real effects when ground truth is available through randomisation.

5.3 Refutations and E-values on the observational effects

Table 5.3. Doubly-robust clause effects on the full 2015 test set with E-values. The three refuters (placebo treatment, random common cause, 80% subsets) passed in every repetition for every clause (pass rate 1.0).

Clause	DR effect (all test loans)	SE	E-value (point)	E-value (CI bound)
dti_gt25	0.0511	0.0047	2.0702	1.9213
fico_lt680	0.0364	0.0037	1.8326	1.7101
lti_gt20	0.0385	0.0035	1.8741	1.7596
util_gt75	0.0090	0.0043	1.3173	1.0597

Passing the refuters rules out gross pipeline failure; it does not prove ignorability. The E-values quantify the remaining assumption: a hidden confounder would need a risk-ratio association of about 1.8-2.1 with both the clause and default to explain away the DTI, FICO and loan-to-income effects (considerable, but not extreme for credit data). The utilization effect is fragile (CI-bound E-value 1.06), consistent with its poor calibration.

5.4 Retriever evidence quality (before and after the fix)

The first full run showed that restricting matching to a text-similarity pool left most evidence sets unbalanced: the median standardised mean difference was 0.49-2.63 and 48-98% of evidence sets were flagged thin. Adding balance-space neighbours across the whole index (v1.1) cut the thin-evidence share to 25-34% and the median SMD to about 0.33-0.41. As a consequence the audit

flag *thin_overlap*, which had dominated all other flags, became a smaller (but still the largest) component.

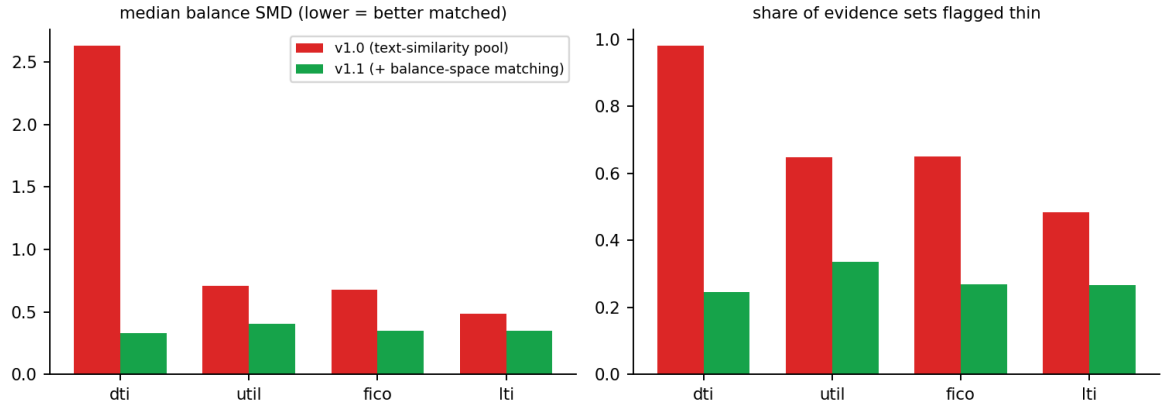


Figure 4. Evidence balance before (red, v1.0) and after (green, v1.1) adding balance-space matching.

Table 5.4. Policy support. Every clause is topically covered but none of the four cutoffs is stated in the retrieved text, so all are labelled analyst-defined in the prompts and the audit. (In v1.0 utilization was matched to the DTI paragraph because the entity pattern was too generous; tightening it fixed that.)

Clause	Best-matching policy passage (v1.2)	Dense score	States the cutoff?
dti_gt25	Fannie Mae B3-6-02 Debt-to-Income Ratios	0.86	no
util_gt75	CFPB 12 CFR 1026.43 (ability to repay)	0.63	no
fico_lt680	Fannie Mae B3-5.1-01 Credit Scores	0.82	no
lti_gt20	CFPB 12 CFR 1026.43 (ability to repay)	0.73	no

5.5 Do language-model systems benefit from causal evidence? (RQ2)

Table 5.5. Discrimination on the same 600 randomly sampled 2015 applicants (85 defaults, 14.2%). Differences vs the plain LLM use a paired bootstrap over applicants; Brier and ECE use cross-fitted Platt scaling of each margin.

System	AUROC	diff vs plain LLM	CI low	CI high	p	Brier	ECE
no_rag	0.633	0.000	0.000	0.000	1.000	0.120	0.027
vanilla_rag	0.626	-0.008	-0.062	0.045	0.768	0.120	0.040
knn_rag	0.639	0.005	-0.039	0.056	0.774	0.119	0.036
causal_rag	0.678	0.044	-0.001	0.089	0.060	0.117	0.027
causal_rag_smooth	0.666	0.033	-0.013	0.080	0.164	0.117	0.025
causal_calf_only	0.658	0.024	-0.014	0.067	0.240	0.118	0.017
causal_precedent_only	0.670	0.036	-0.001	0.074	0.062	0.117	0.038
causal_conditions_only	0.668	0.035	-0.001	0.071	0.068	0.117	0.056
causal_graded	0.660	0.027	-0.023	0.078	0.258	0.117	0.035
bands_only	0.647	0.013	-0.008	0.036	0.222	0.118	0.029
causal_graded_bands	0.655	0.022	-0.026	0.075	0.360	0.118	0.042
CALF factual risk (tabular)	0.707	0.073	0.008	0.139	0.028	0.114	0.028
LightGBM (tabular ceiling)	0.691	0.057	-0.011	0.127	0.106	0.115	0.030

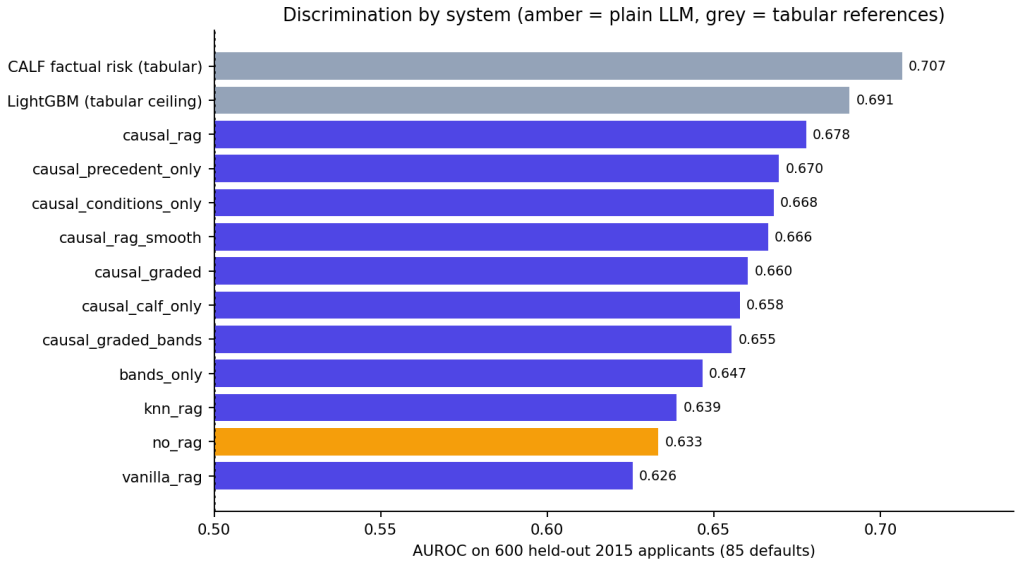


Figure 5. AUROC by system on 600 held-out applicants.

- **No LLM variant is significantly better than the plain LLM.** The best, Causal-RAG, reaches 0.678 (+0.044, p = 0.060); vanilla and kNN RAG are not better than no retrieval at all (0.625, 0.639).
- **Tabular models are at least as good as every LLM system**: CALF factual risk 0.707 (+0.073 over the plain LLM, p = 0.028) and LightGBM 0.691. The project's claim is therefore about causal consistency and auditability, not about beating tabular models.
- **The first reading was optimistic (winner's curse).** On the first 300 applicants Causal-RAG scored 0.707-0.710; on the full 600 it is 0.678. With only 42 defaults, AUROC differences below about 0.07 were pure noise; the larger sample (85 defaults) shrank the apparent gain.

Table 5.6. The Causal-RAG result across runs and sample sizes.

Run	Applicants	Defaults	Causal-RAG AUROC	Plain LLM AUROC
v1.0 (old retrieval/policy)	300	42	0.641	0.645
v1.1 (fixed retrieval, dti_gt25)	300	42	0.707	0.645
v1.2 (first 300 of the 600)	300	42	0.710	0.645
v1.2 (all 600)	600	85	0.678	0.633

Control experiments

Table 5.7. Paired-bootstrap comparisons that decide what actually drives the result.

System A	System B	AUROC A	AUROC B	A - B	CI low	CI high	p
causal_conditions_only	no_rag	0.668	0.633	0.035	-0.001	0.071	0.068
causal_rag	causal_conditions_only	0.678	0.668	0.010	-0.028	0.048	0.594
causal_precedent_only	causal_conditions_only	0.670	0.668	0.002	-0.035	0.034	0.908
causal_calf_only	causal_conditions_only	0.658	0.668	-0.010	-0.041	0.023	0.546
causal_rag	causal_calf_only	0.678	0.658	0.020	-0.006	0.045	0.126
causal_rag	causal_precedent_only	0.678	0.670	0.008	-0.023	0.041	0.614
causal_graded	causal_conditions_only	0.660	0.668	-0.008	-0.053	0.041	0.796
causal_graded	causal_rag_smooth	0.660	0.666	-0.006	-0.041	0.026	0.758
bands_only	no_rag	0.647	0.633	0.013	-0.008	0.036	0.222
causal_graded_bands	no_rag	0.655	0.633	0.022	-0.026	0.075	0.360
causal_graded_bands	causal_graded	0.655	0.660	-0.005	-0.019	0.010	0.458
causal_graded_bands	bands_only	0.655	0.647	0.009	-0.037	0.060	0.684
causal_graded_bands	causal_rag	0.655	0.678	-0.022	-0.070	0.028	0.416

- **The conditions-only control matches the full pipeline** (0.668 vs 0.678, p = 0.59). Telling the LLM which standard risk flags apply is enough to reproduce the gain; the matched precedent (0.670)

and the CALF effect numbers (0.658) each add nothing measurable on top.

- **Band tables did not help** (bands-only 0.647, graded + bands 0.655; neither distinguishable from the plain LLM).
- The most likely explanation is a feature-engineering effect: the flags (DTI above 25%, FICO below 680, ...) are real risk indicators the plain LLM does not compute on its own. This is a hypothesis; the experiments rule out the causal evidence as the driver but do not by themselves prove the flag explanation.

5.6 Is the LLM consistent with the causal estimates? (RQ3)

Table 5.8. Counterfactual consistency over 2,090 material regime-flip pairs (clauses with |CALF effect| of at least 1 percentage point). Response rate = share where P(decline) moved by at least 0.005. Evidence response = share of cases where reversing the matched-precedent counts moved the decision; spurious flip rate = share of decisions crossing the threshold when only state / employer / loan title changed.

System	respons e rate	sign agree (all)	sign agree (responder s)	"all clauses raise risk" baseline	Spearma n dP vs tau	evidence response	evidence correct sign	spurious flip rate
causal_rag	0.443	0.820	0.980	0.680	0.023	0.121	0.945	0.053
causal_rag_smooth	0.481	0.921	0.989	0.680	-0.089	0.136	0.939	0.033
no_rag	0.192	0.807	0.863	0.680	-0.264			0.047
vanilla_rag	0.125	0.734	0.737	0.680	-0.174			0.043
causal_conditions_only	0.506	0.892	0.958	0.680	-0.139			0.033
causal_graded	0.643	0.981	0.986	0.680	0.018	0.368	0.964	0.030

- **Sign agreement looks high but is an easy test.** Nearly every clause raises default risk, so a rule that always assumes "a clause raises risk" scores 0.68. The plain LLM already scores 0.81. Smooth (0.92) and graded (0.98) exceed it mainly because they read CALF's numbers.
- **Size of response does not track size of effect.** The Spearman correlation between the LLM's response and CALF's effect size is near zero or negative for every system.
- **The LLM largely ignores matched-precedent counts** in the v1.1 and smooth prompts (it moves in only 12-14% of reversals), although it moves in the right direction when it does (94-95%). The graded prompt made the model use the evidence far more (37% of reversals, 96% correct direction).
- **Irrelevant edits** flip 3-5% of decisions for every system (about 18-32 of 600 cases; the differences between systems are not statistically solid).

5.7 The audit agent (RQ4)

Table 5.9. Decision mix and realised default rates by audit decision. The declined-group rate is derived from the totals. Group sizes are small (declined n = 27-95), so treat the rates with care.

Run	n	approv e	decline	refer	default | approved	default | referred	default | declined (derived)	default if refers were approved
v1.0	300	0.637	0.090	0.273	0.094	0.244	0.148	0.139
v1.1	300	0.673	0.110	0.217	0.094	0.185	0.333	0.116
v1.2 (smooth prompt)	600	0.710	0.158	0.132	0.115	0.177	0.232	0.125

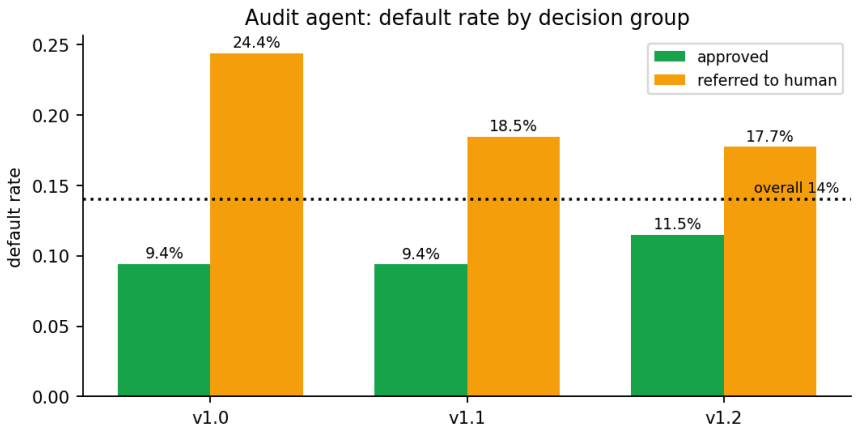


Figure 6. Realised default rate among approved and referred cases by run (dotted line = overall 14%).

- In the final audited configuration (v1.2, n = 600) the ordering is monotone and sensible: approved 11.5%, referred 17.7%, declined about 23% default. Approving the referred cases as well would raise the approved-group default rate from 11.5% to 12.5%, so referral removes about one percentage point of risk.
- Flag counts (v1.2): thin_overlap 162, spurious_sensitive 39, evidence_model_conflict 8, evidence_ignored 5, llm_calf_disagree 4. Most referrals reflect weak evidence rather than detected inconsistency.
- An example rationale generated for an audited case: "*While the applicant's credit score is below the recommended threshold (Condition 3), the estimated increase in default probability is relatively low at +4.8 percentage points. The other conditions are not triggered, and the requested loan amount is well within acceptable limits. Therefore, approving the loan is justified.*" The rationale cites only conditions and numbers supplied by the audit table; the table, not the prose, is the record.

5.8 The cliff test: does the LLM react to the cutoff itself?

A mistake in our own metric. Runs v1.1-v1.3 used a "boundary flip" that moves a field to just across its cutoff *from wherever it currently is*. For an applicant far from the cutoff that is a large move, not a half-point nudge, so the resulting "cliff ratio" (about 1.0 for the smooth prompt) was confounded by move size. Comparisons across systems were still valid (same edits), but the absolute reading was wrong. The local cliff test (Section 4.10) replaced it, and it reversed our earlier conclusion about the smooth prompt.

Table 5.10. Local cliff test (150 applicants x 4 clauses; the number of valid pairs per system is shown; bootstrap over applicants). 1 = smooth.

Run	System	pairs	mean |change| across cutoff	mean |change| placebo	cliff index	CI low	CI high	reading
v2.0	bands_only	595	0.102	0.139	0.734	0.628	0.841	smooth
v1.4	no_rag	595	0.043	0.040	1.085	0.807	1.493	smooth
v2.0	causal_graded_bands	595	0.136	0.059	2.320	1.888	2.897	mild cliff
v1.4	causal_graded	595	0.137	0.050	2.763	2.213	3.464	mild cliff
v1.4	causal_conditions_only	595	0.296	0.056	5.285	4.002	6.998	cliff
v1.4	causal_rag_smooth	595	0.327	0.004	86.037	53.303	172.936	cliff

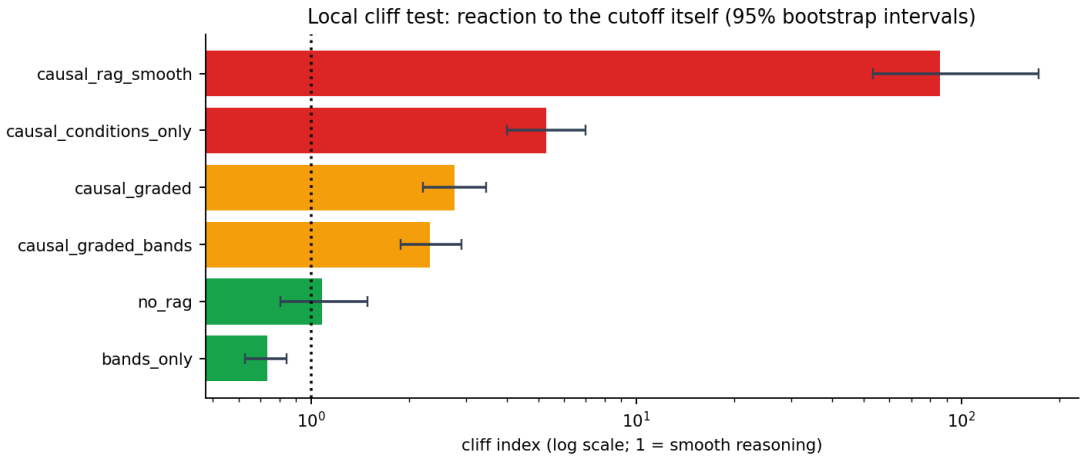


Figure 7. Cliff index by system (log scale). Green = smooth, amber = mild cliff, red = cliff.

Table 5.11. Cliff index by clause (values for the smooth prompt are huge because its output is almost flat away from the cutoff, so the ratio's denominator is close to zero).

Clause	no_rag	bands_only	graded	graded + bands	flags only	smooth (label)
dti_gt25	1.66	1.31	3.50	2.77	14.42	35.44
fico_lt680	0.83	0.88	0.90	0.80	2.04	6882.86
lti_gt20	2.35	1.21	4.17	4.12	24.78	960.32
util_gt75	1.11	0.49	6.49	3.95	2.78	56.16

- **The plain LLM is smooth** (1.08 [0.81, 1.49]); so is the prompt that shows only band tables and no cutoff wording (0.73 [0.63, 0.84]).
- **The smooth prompt, designed to remove the cliff, is the most cliff-like system** (86 [53, 173]): its answer is nearly flat until the TRIGGERED label flips, then jumps by about 0.33 in P(decline). Flags-only is also a cliff (5.3).
- **The graded prompt (no TRIGGERED word) cuts the cliff to 2.8, and graded + bands to 2.3**, but both intervals exclude 1: a mild cliff remains. It is worst for loan-to-income (about 4) and utilization (about 4-6), and absent for FICO (0.8-0.9).
- **Interpretation.** Every system whose prompt describes conditions in terms of cutoffs or regimes reacts to the cutoff itself; the two systems without such language are smooth. Adding band tables did not remove the mild residual cliff. These data cannot separate the cutoff wording from the precedent text and CALF numbers that always come with it.
- **Why it matters.** A step at a conventional threshold is exactly the hard-rule behaviour an auditor wants to detect: real risk changes continuously across 25% DTI, but the label-driven prompts treat 24.9% and 25.1% very differently.

5.9 Band tables used in the final experiment

Table 5.12. Unadjusted default rate by value band among the 306,212 historical loans in the retrieval index (the same table is shown to every applicant in the band prompts). Risk rises monotonically with DTI, utilization and loan-to-income and falls with FICO; adjacent bands differ by only about 1-5 percentage points of default rate, far smaller than the 0.33 jump in P(decline) seen in the label prompts.

Field	Bands and unadjusted default rate
dti	<10%: 10.3%; 10-15%: 11.3%; 15-20%: 13.1%; 20-25%: 14.9%; 25-30%: 17.0%; 30-35%: 18.0%; 35%+: 22.7%
revol_util	<25%: 10.0%; 25-50%: 12.3%; 50-75%: 13.8%; 75-90%: 14.7%; 90%+: 16.0%
fico	<680: 17.2%; 680-699: 14.1%; 700-719: 10.7%; 720-739: 8.3%; 740+: 5.6%
lti	<10%: 10.6%; 10-15%: 11.5%; 15-20%: 12.3%; 20-25%: 13.1%; 25%+: 16.6%

6. Problems we met and how we solved them

Real data and real infrastructure produced many problems. They are listed here in full, because several of them changed the conclusions. Items 1-16 are methodological; items E1-E8 are engineering and environment problems.

#	Problem	Solution / outcome
1	Real data contain no ground-truth individual effects, so PEHE cannot be computed.	Four kinds of evidence instead: doubly-robust pseudo-outcomes (DR-MSE, BLP slope, quintile tables), a real randomised experiment (Hillstrom), refutation tests, and E-values.
2	The paper selects checkpoints by validation PEHE (needs true effects).	Checkpoints are selected by MSE against a cross-fitted DR pseudo-outcome on the validation split.
3	Leakage: the lender's own grade, sub-grade and interest rate, and payment/recovery columns, encode the old decision or the outcome.	Only pre-origination columns are kept; grade, sub_grade, int_rate are excluded as mediators of the previous decision.
4	If the column that defines a clause stays in the inputs, the treatment is a function of the covariates and overlap collapses.	Per-clause drop lists remove the defining columns (and mechanically tied ones); the graph adds a second, stricter adjustment set for sensitivity.
5	The Fannie Mae 36% DTI clause binds on only 1.5% of Lending Club loans: 85% of test loans had propensity below 0.05 and CALF's slope was -0.16.	Redefined as dti_gt25 (20.5% treated), explicitly labelled analyst-defined; the rarity of the 36% cutoff is reported as a finding.
6	The paper's cross-attention (one vector per layer) has a single key, so the softmax is identically 1: it is a linear map, not selective attention.	Added k_tokens (4 latent tokens per layer); k_tokens = 1 reproduces the paper. Both were run: no measurable difference on these data.
7	A built-in "effects are non-negative" prior would make the counterfactual audit circular.	Monotonicity weight set to 0 by default.
8	MC-Dropout intervals under-cover (66.9% vs 90% in the paper).	Seed ensembles with between-seed spread; MC-Dropout kept only as an option and flagged uncalibrated.
9	"Policy-supported" only meant topically related: the utilization clause was matched to a DTI paragraph and every clause showed as supported although no cutoff was stated.	*supported* now requires the retrieved text to state the cutoff (regex); otherwise the prompt and audit say "analyst-defined". The utilization entity pattern was tightened (v1.2 maps it to the ability-to-repay rule).
10	Matched evidence was thin or unbalanced: median SMD 0.49-2.63, 48-98% of sets flagged thin, so the audit flag thin_overlap dominated.	Added balance-space nearest neighbours across the whole index to the candidate pool and made balance dominate the distance: thin share 25-34%, SMD about 0.33-0.41.
11	With 300 applicants (42 defaults) Causal-RAG looked significantly better (AUROC 0.707, p = 0.046); at 600 applicants it was 0.678 (p = 0.06).	Sample doubled, intervals reported everywhere, a "first 300" table kept for like-for-like comparison, and the result described as a winner's curse.
12	Sign agreement with CALF looked excellent (0.92-0.98) but is an easy test: nearly every clause raises risk.	Added the "every clause raises risk" baseline (0.68), the Spearman correlation of effect sizes (near 0) and a within-clause correlation; sign agreement is no longer used alone.
13	CALF's outputs might leak outcome information into the LLM's prompt and inflate its AUROC.	The LLM sees effect estimates but never CALF's factual risk; tabular CALF/LightGBM are separate reference rows; control experiments isolate each prompt ingredient.
14	Was the improvement caused by the causal evidence or just by the flags?	Control systems: flags-only, CALF-only, precedent-only. Flags-only matched the full pipeline (0.668 vs 0.678, p = 0.59): the causal evidence added nothing measurable.
15	Our "boundary flip" cliff metric moved the field to just across the cutoff from wherever it was, so it was confounded by move size; the smooth prompt looked cliff-free (ratio about 1.0).	Replaced by the local cliff test (equal-width placebo interval, bootstrap intervals). It showed the smooth prompt is the cliffiest system (index 86); the earlier reading is retracted.
16	The graded prompt (no TRIGGERED word) and band tables only partly removed the cliff (2.8 and 2.3 vs 1.08 for the plain LLM).	Reported as a mild residual cliff tied to cutoff/regime wording; a bands-only prompt without such wording is smooth (0.73). Disentangling wording from evidence content remains future work.

Engineering and environment problems

#	Problem	Solution / outcome
E1	EconML causal-forest baseline crashed (a classifier was used for a binary outcome).	Fixed (regressor for the outcome model); it ran later but is off by default because it is very slow.
E2	The Criteo uplift download link (scikit-uplift) returns HTTP 404.	Skipped gracefully; Criteo is optional and Hillstrom is the validation experiment.

#	Problem	Solution / outcome
E3	The Hillstrom loader accepted a results file whose name contained "hillstrom" and crashed (no *segment* column).	The loader validates required columns for every candidate file, falls back to the original site and scikit-uplift, and the RCT step is non-fatal.
E4	A notebook session without a working GPU took 367 min for embeddings (4.8 min on GPU) and 146 min for CALF (21 min), about 12 hours for 5 stages.	The setup cell prints the GPU and refuses to start long stages on CPU; stages 03-04 skip themselves when their outputs exist.
E5	The notebook imported an old or incomplete copy of the code from the attached dataset (ImportError, missing function).	The setup cell lists every copy, picks a complete and current one (checked by file presence and a feature marker), clears stale imports and gives an explicit upload instruction.
E6	Results were lost when interactive sessions ended; the restore function copied 0 files because Kaggle mounts inputs deeper (/kaggle/input/datasets/user/name/...).	Recursive restore that ignores archived results/ folders; notebook outputs can be attached directly as inputs; a packaging cell zips the artifacts; run history is archived in the code dataset.
E7	Long LLM stages risked being lost to timeouts.	Chunked, resumable stages (100 applicants per chunk; 25 for the cliff test); every run exports its tables for the next comparison.
E8	GPU daily quota and 12-hour session limits.	Runs were scheduled as Commit versions sized to fit (LLM stages 83-351 min); the final run needed 143 min.

7. Run history

Version	What changed	Applicants / defaults	Key result	LLM stage time (T4)
v1.0	First full GPU run. dti_gt36, topical-only policy support, text-pool matching; EconML crashed.	300 / 42	Causal-RAG 0.641 (no gain); evidence thin in 48-98%	83 min LLM stage (about 2 h 6 min total)
v1.1 (CPU session)	Stages 1-5 only, run without a working GPU.	-	Retriever balance and thin-evidence fixes confirmed (Section 5.4)	12 h (stage 03: 367 min, stage 04: 146 min)
v1.1	dti_gt25, strict policy cutoff matching, balance-space matching, new diagnostics.	300 / 42	Causal-RAG 0.707 (+0.062, p = 0.046); audit decisions ordered	94 min
v1.2	Smooth prompt, CALF-only and precedent-only controls, resumable chunks, 600 applicants.	600 / 85	Causal-RAG 0.678 (+0.044, p = 0.06); smooth 0.666; controls 0.658-0.670	351 min
v1.3	Flags-only control, graded prompt, boundary flips for the baselines.	600 / 85	Flags-only 0.668 matches Causal-RAG; graded 0.660; evidence use 37% for graded	295 min
v1.4	Local cliff test (correction of the boundary-flip metric).	150 applicants x 4 clauses	Cliff index: plain 1.08, graded 2.76, flags 5.29, smooth 86	125 min
v2.0	Band-table prompts and the automatic final report.	600 / 85 and 150	Bands-only 0.647, smooth (0.73); graded + bands 0.655, mild cliff 2.32	143 min

8. Conclusions

8.1 Answers to the research questions

- **RQ1 (can CALF estimate clause effects credibly on real loans?): yes, with caveats.** Heterogeneity is well calibrated for FICO, loan-to-income and DTI, validated against a real randomised experiment, and robust to the refuters. Utilization is weak. Layering added nothing over a flat network, in line with the paper's own insurance result. Effects remain observational: ignorability is untestable (E-values about 1.8-2.1) and the data contain only funded loans.
- **RQ2 (does causal evidence improve the LLM's decisions?): not measurably.** The best variant gained +0.04 AUROC (p = 0.06), a flags-only control reproduced it, and tabular models were at least as good. The most plausible explanation is that listing standard risk flags helps; the causal evidence adds nothing detectable on 85 defaults.
- **RQ3 (is the LLM consistent with the causal estimates?): only in direction.** It moves the right way almost always, but that is an easy test; its response size does not track the effect size, it largely ignores the matched-precedent counts (except in the graded prompt), and prompts that name

cutoffs make its behaviour step-like (cliff index 2.3 to 86 against 1.08 for plain reasoning).

- **RQ4 (can it be made auditable?): yes, with modest benefit.** Each case yields a decision, flags, a clause table, evidence ids and a causal DAG; decisions are monotone in realised risk (approved 11.5%, referred 17.7%, declined about 23%). Most referrals reflect weak evidence rather than detected inconsistency.

8.2 What the project contributes

- A complete, reproducible, real-data pipeline that applies CALF to a banking problem and validates it without ground-truth individual effects.
- A set of audit tools that proved their worth by catching our own errors: the control experiments (flag-listing effect), the "easy sign test" baseline, the winner's-curse check and the corrected cliff test.
- Concrete evidence about how prompt wording shapes behaviour: cutoff and regime language induces step-like decisions, number-only presentations stay smooth.
- A negative-result-honest record of seven runs, archived with their tables so that nothing is lost and new runs can be compared automatically.

8.3 Limitations

- Effects are observational and depend on the causal graph; funded loans only (selection bias; no reject inference); clauses are group contrasts, not nudges of a continuous value.
- The four cutoffs are analyst-defined (the retrieved policy text states none of them) and Fannie Mae's 36% DTI cutoff binds on only 1.5% of these loans.
- LLM evaluation uses 600 applicants with 85 defaults: AUROC differences below about 0.05 are within noise, and the cliff test uses 150 applicants.
- One LLM (Qwen2.5-7B, 4-bit) and one prompt family; other models may behave differently. Results come from single runs per configuration.
- The cliff test cannot separate cutoff wording from the evidence content that accompanies it; the live demo has not been run on Kaggle with the real model; and nothing here is a lending decision system.

8.4 Suggested future work

- Dose-response prompts that give the matched-precedent evidence by value band (not by regime) to see whether the residual cliff disappears while keeping the causal content.
- A larger evaluation sample (more defaults) and several LLMs, with fine-tuning or calibration of the decision threshold per system.
- Apply CALF where the covariate layers are genuinely hierarchical (the paper's NLSY97 setting) to test the structure hypothesis again.
- Reject inference and conformal recalibration of CALF's uncertainty; sensitivity analyses beyond E-values.

9. How to reproduce and use the project

- **Code dataset.** Upload `causal_rag_kaggle.zip` as a Kaggle dataset (it contains the package, notebooks, results of all runs, README, RESULTS_INDEX and FINAL_SUMMARY). Attach only one copy.
- **Notebooks.** `FINAL_v2_0_last_run.ipynb` runs the last experiment and builds the automatic report (about 1.5-2 h on a T4); `LIVE_DEMO.ipynb` launches the Gradio demo; older stage notebooks live in `notebooks/archive_earlier_versions/` for reproducing earlier runs.
- **Artifacts.** A run's `artifacts/` folder (cleaned loans, CALF models and scores, indexes, evidence bundles, LLM scores) can be attached to the next notebook directly from the previous notebook's Output tab, or saved as a dataset; stages skip themselves when their outputs exist.
- **Run discipline.** First run with SMOKE = True (a dry run on a small real sample), then SMOKE = False as a Commit (Save & Run All). The setup cell checks the GPU and the code version. Long stages are resumable.
- **Outputs.** Each run exports tables, logs and (for the final run) FINAL_REPORT.html/.md with figures; copy the export folder into the dataset's `results/` directory to keep the run in the history.

Archived run folder (results/)	Contents
v1_0_first_full_run	CALF results, RCT, refutations, retriever log, discrimination, counterfactual table, coverage
v1_1_stage01_05_cpu_session	data prep, graph, policy index logs, graph image, timings, retriever stats
v1_1_full_run	all stages; CALF, RCT, refutations, discrimination, counterfactual, coverage
v1_2_run	600 applicants, controls, pairwise tests, coverage, RCT quintiles
v1_3_run	flags-only and graded systems, boundary flips for baselines
v1_4_cliff_run	local cliff table and by-clause table
v2_0_final	band prompts: discrimination, pairwise, cliff tables, logs

Appendix A. Glossary

Term	Meaning
ATE / CATE / ITE	Average / conditional average / individual treatment effect: the change in outcome caused by the treatment, overall / for a profile / for one unit.
Clause / regime	A binary underwriting condition (e.g. DTI above 25%); "regime" = being inside it rather than outside it.
Overlap / positivity	Every kind of applicant has a chance of being both inside and outside the regime; violated when propensity is near 0 or 1.
Propensity score e(x)	Probability of being inside the regime given background x.
Doubly-robust (AIPW) pseudo-outcome	A noisy but unbiased signal of the individual effect that is correct if either the propensity or the outcome model is correct; used to evaluate models without ground truth.
DR-MSE	Squared error between predicted effects and the DR pseudo-outcome; ranks models like PEHE would.
BLP slope (calibration)	Slope of the DR pseudo-outcome on the predicted effect: 1 = well scaled, 0 = no signal, between 0 and 1 = over-dispersed.
GATES	Group average treatment effects by predicted-effect quantile; observed should rise with predicted.
Qini	Area between a model's uplift curve and random targeting, computed from real randomised outcomes.
SMD	Standardised mean difference: balance between matched treated and control groups (below about 0.1 is good, above 0.5 is poor).
E-value	Strength of association (risk-ratio scale) an unmeasured confounder would need with both treatment and outcome to explain an effect away.
AUROC	Probability that a random defaulter is scored riskier than a random non-defaulter (0.5 = chance).
Paired bootstrap	Resampling applicants to get an interval for the difference between two systems scored on the same applicants.
Platt scaling / Brier / ECE	Calibration of scores to probabilities; mean squared probability error; expected calibration error.
Regime flip / boundary flip / local cliff test	Counterfactual edits of an applicant: move to the typical value of the opposite regime / just across the cutoff / compare the change across the cutoff with an equal-width interval away from it.
Cliff index	Change in P(decline) across the cutoff divided by the change over an equal-width interval elsewhere; 1 = smooth.
Winner's curse	The best-looking result in a small sample tends to be an overestimate and shrinks on more data.

Appendix B. Observations on the CALF paper

While reading the paper for this project we noticed several internal inconsistencies worth correcting before submission:

- The true ATE is stated as $48,368.63 in Section V but as 48,734 in Section VI.
- The Jobs benchmark "ATE error" column does not equal |predicted - true| (e.g. T-Learner: 2455.06 - 1833.18 is about 622, not 25.33), and the text refers to a note ("mislabeled scale corrected: see note") that does not exist.
- On IHDP the Causal Forest ATE error is shown as 2.278 although |3.956 - 4.05| is about 0.09.
- The cross-attention in Eqs. 3-5 is applied to a single vector per layer, which makes the softmax trivial (see Problem 6); the 4.3% ablation difference attributed to attention may therefore reflect an extra linear map rather than selective attention.
- The paper itself reports that the structure helps only when covariates decompose into causal layers (its insurance result); our Lending Club and Hillstrom results agree.

Appendix C. Prompt examples

Common ending for every system (the answer is read from the logits of the letters A and B):

```
Should this loan be approved?
```

```
A = Approve (expected to be repaid in full)
B = Decline (expected to default)
Answer with one letter only.
```

Flags-only control (after the application card):

```
STANDARD RISK FLAGS TRIGGERED
- Revolving credit utilization above 75%
- Requested amount above 20% of annual income
```

Bands-only control (after the application card; values from the real retrieval index):

```
HISTORICAL DEFAULT RATES BY BAND (all past 36-month loans 2012-2014, unadjusted for other factors)
- DTI: <10% 10.3%; 10-15% 11.3%; 15-20% 13.1%; 20-25% 14.9%; 25-30% 17.0%; 30-35% 18.0%; 35%+ 22.7%   (each with n=...)
- FICO: <680 17.2%; 680-699 14.1%; 700-719 10.7%; 720-739 8.3%; 740+ 5.6%
```

Graded Causal-RAG block (one of four conditions; every applicant sees all four in the same layout, and there is no TRIGGERED word):

```
Condition 3: Credit score below 680. Applicant: FICO 677 - 3.0 points below the cutoff (regime boundary < 680).
  Policy context (topic only; the policy text does not state this cutoff, so the threshold is analyst-defined) [Fannie Mae
  Matched precedent (past borrowers compared like-for-like on background factors): inside this regime a of n defaulted (x%)
  outside it b of m defaulted (y%). Difference +z pp (95% CI ...).
  Estimated effect on THIS applicant's default probability of being inside this regime rather than outside it: +4.8 pp (mod
The cutoffs are analyst-defined conventions, not cliffs: an applicant just inside and just outside a cutoff has almost the
```